

⚡ Production-Ready RAG Knowledge Extraction System

Final Project Presentation & System Demonstration

Author: Raheela Daud (AI Intern) | **Organization:** Parallax Labs
Internship | **Project:** End-to-End RAG Engine, NLP Enrichment &
FastAPI Service

Slide 1: Problem Statement & Project Scope

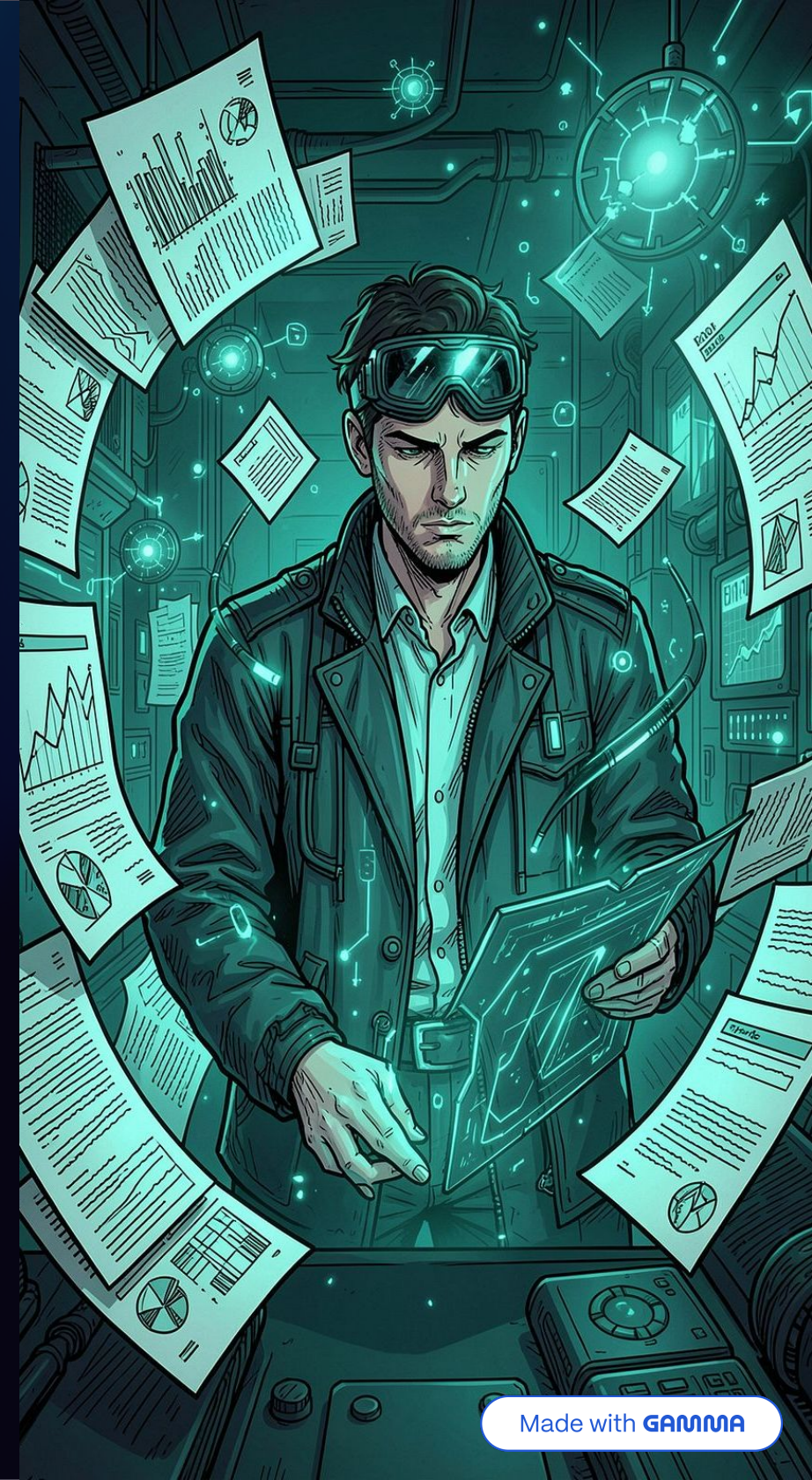
Challenge

Traditional LLMs suffer from domain hallucinations, lack grounding on domain-specific computer science data, and incur high API token costs.

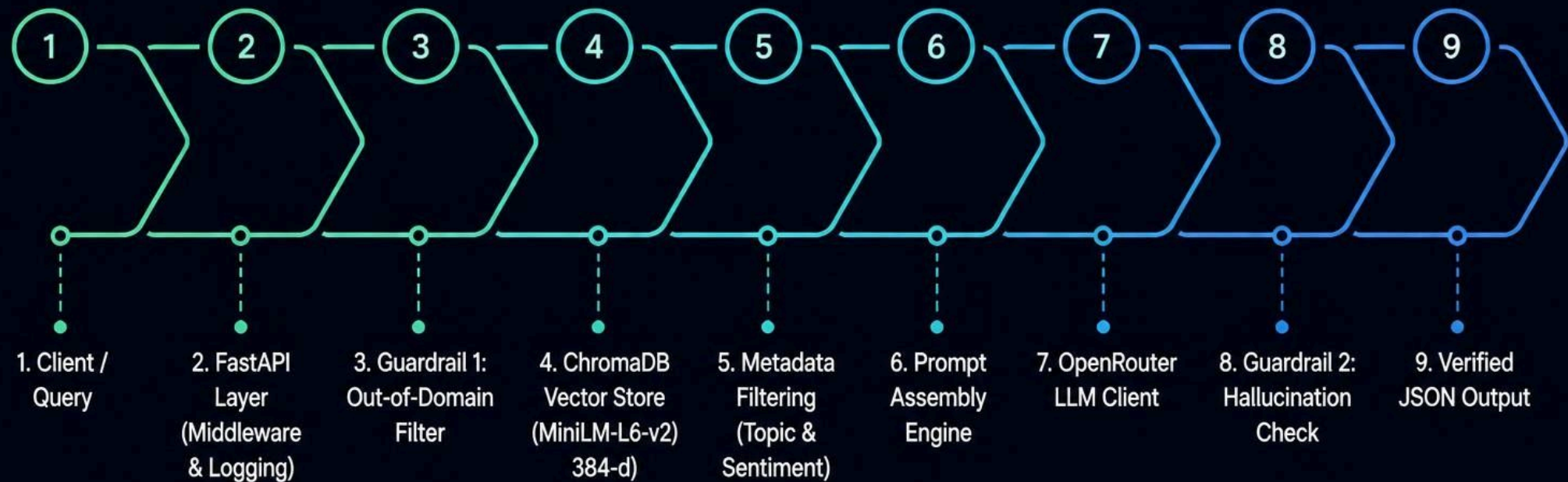
Objective

Build an end-to-end, reproducible RAG pipeline over 5,100 technical documents with:

- Strict grounding and hallucination guardrails.
- NLP Metadata Enrichment (LDA Topic Modeling + Lexicon Sentiment).
- Sub-second to low-latency vector retrieval via ChromaDB.
- Production REST API layer deployed with FastAPI and Pytest validation.



Slide 2: End-to-End Technical Architecture



Slide 3: NLP Topic Modeling & Sentiment Enrichment (Week 4)

Topic Modeling

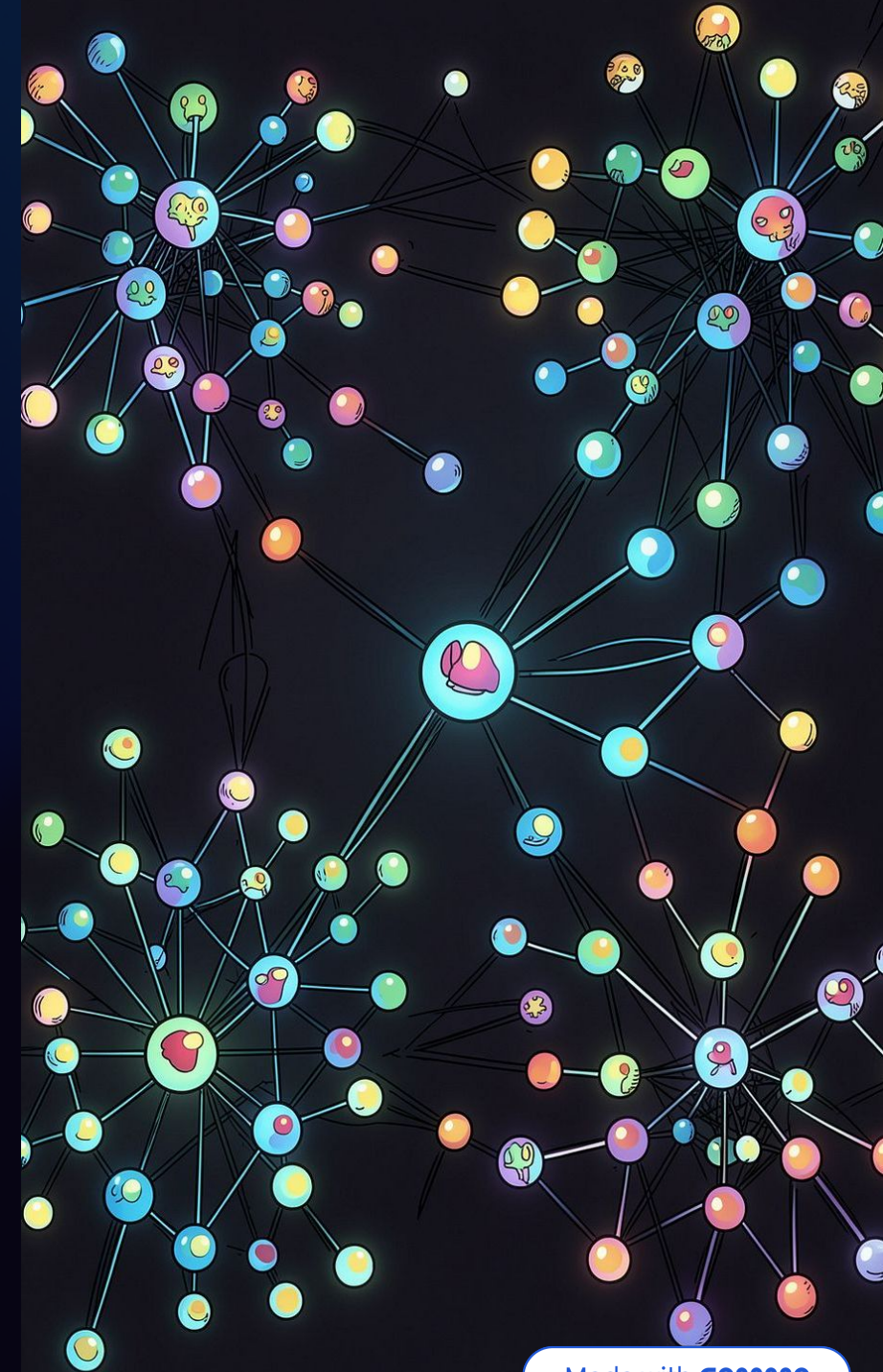
Latent Dirichlet Allocation (LDA) discovered 5 latent topic clusters across technical domains (Machine Learning, Systems, Algorithms, Networks, Security).

Sentiment Analysis

Custom lexicon analyzer scoring document sentiments into Positive, Neutral, and Negative.

Metadata-Enriched Search

Vector database queries filtered by topic and sentiment tags to enhance retrieval precision and context relevance.



Slide 4: Key Evaluation Benchmarks & Metrics (Week 5 & 6)

Evaluation Metric	Benchmark Score	Standard / Methodology
Retrieval Precision@3	100.00%	Ground-truth keyword verification
Retrieval Recall@3	100.00%	Relevant document intersection
Sentiment Validation Accuracy	83.33%	Labeled technical benchmark set
Average End-to-End Latency	~1200 ms - 4000 ms	Full retrieval + OpenRouter generation
Out-of-Domain Interception	100% Intercepted	Safety guardrails blocking off-topic inputs
FastAPI Unit Test Pass Rate	100% (5/5 Passed)	Pytest test suite with TestClient



Slide 5: Production API & Endpoint Specification

GET /health

Service health status verification.

POST /query

End-to-end grounded RAG execution with latency tracking.

POST /search/filtered

ChromaDB vector retrieval with topic/sentiment constraints.

Interactive Swagger UI

Accessible live at `/docs`.

Slide 6: Summary & Future Scope

Key Achievements

Delivered a fully documented, tested, reproducible RAG system meeting all 6-week milestones.

Future Work

Integration of Hybrid Search (BM25 + Dense Vectors) and Cross-Encoder Re-ranking for enhanced retrieval accuracy.